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Microscopic Description of Trapped Hard Rods in One Dimension

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Notation and abbreviations

Symbol	Meaning
a	Rod diameter, also the minimum distance between particle centers.
a_{ho}	Oscillator length, $a_{\text{ho}} = \sqrt{\hbar/(m\omega)}$.
α	Variational width parameter of the trial wave function at nonzero rod diameter.
$b_{\text{fit}}, b_{\text{range}}$	Bounds from the choice of fit and the range of fitted time steps.
$b_M, b_{\tau M}$	Bounds from the effect of the finite walker population and its dependence on the time step; Appendix B.3.
$\Delta\tau$	Time step in imaginary time; $\Delta\tau\omega$ is dimensionless.
E_T	DMC reference energy.
ℓ	Forward time.
M	Walker population.
N	Number of hard rods.
$n(x)$	One-body density, normalized to N .
Q	Center of mass coordinate, $Q = N^{-1} \sum_i q_i$.
q	Position in oscillator units, $q = x/a_{\text{ho}}$.
q_{max}	Positive endpoint of the density grid $[-q_{\text{max}}, q_{\text{max}}]$.
R_{rms}	RMS radius of the cloud.
$\tau_{\text{heat}}, \tau_{\text{prod}}$	Equilibration and sampling times in the DMC calculation.
u_i	Reduced ordered coordinate, defined in Eq. (4.1).
Y	Center of mass in reduced coordinates, $Y = N^{-1} \sum_i y_i$.

Abbreviation	Meaning
DMC	Diffusion Monte Carlo.
ESJD	Expected squared jumping distance.
ESS	Effective sample size.
FW	Forward walking.
LDA	Local density approximation.
MALA	Metropolis-adjusted Langevin algorithm.
MCSE	Monte Carlo standard error.
RMS	Root mean square.
RWM	Random-walk Metropolis.
TG	Tonks–Girardeau.
VMC	Variational Monte Carlo.

Abstract

This thesis studies a one-dimensional system of quantum hard rods in a harmonic trap. Systems with $N = 10$ and 20 particles are considered, from the hard point limit, $a = 0$, up to a rod diameter of $a/a_{ho} = 10$. The ground-state properties are calculated with variational Monte Carlo (VMC) and diffusion Monte Carlo (DMC) methods, and then compared with the predictions of the local density approximation (LDA). The LDA calculation uses the homogeneous hard-rod equation of state, which accounts for excluded volume and is known analytically.

The VMC method is based on sampling the trial wave function, and the implementation is checked by ensuring that the two kinetic-energy estimators are consistent. The DMC method relies on importance sampling and gives the ground-state energy. The density profile and root mean square (RMS) radius are obtained by the forward-walking method. Calculations were repeated at different time steps, walker populations, and forward times to check the numerical accuracy of these results.

It was found that for finite rod diameters, the DMC energy and cloud size are close to the LDA values. At fixed rod diameter, the agreement improves from $N = 10$ to $N = 20$. At the same time, the DMC and LDA density profiles show a clear difference. The Friedel oscillations observed in the hard point limit become stronger as the rod diameter increases. At $a/a_{ho} = 10$, the density profile contains N narrow peaks, whereas the smooth LDA density profile describes the overall envelope. Averaging one of the microscopic profiles over cells that each contain one particle recovers the LDA curve near the center of the trap.

Keywords: one-dimensional hard rods; diffusion Monte Carlo; variational Monte Carlo; forward walking; local density approximation.

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1 Introduction

The system studied in this thesis is composed of quantum particles that are trapped in a harmonic oscillator and are confined to a one-dimensional geometry. The particles in this system behave as hard spheres or, as they are called in one dimension, hard rods of diameter a . Two hard rods cannot get closer than their diameter, and it is not allowed for the particles to pass through one another.

When the harmonic trap is absent, the homogeneous hard-rod system is integrable. Correspondingly, its ground-state energy is known exactly. Adding a harmonic trap causes the system to lose its integrability and exact results are not known for $N > 2$ at nonzero rod diameter. Thus, the goal of this thesis is to calculate the ground-state energy, density profile, and root mean square (RMS) radius with variational Monte Carlo (VMC) and diffusion Monte Carlo (DMC) methods and compare them with the local density approximation (LDA). LDA applies the homogeneous hard-rod equation of state at each point in the trap. It is quick to calculate and gives a smooth density profile, but it cannot reproduce the N separate maxima of a finite cloud.

This contrast is the main reason for the comparison. The energy and size of the cloud can remain close to LDA even when the microscopic density has a different shape. Eight systems are studied, with $N = 10$ and 20 and four values of the rod diameter.

1.1 Objectives

The objectives of this study are as follows:

1. implement the VMC method for one-dimensional trapped hard-rod systems;
2. compare different estimators for the kinetic energy within VMC approach;
3. implement the DMC method with importance sampling;
4. verify that the DMC algorithm without branching reproduces the VMC distribution and observables;
5. study different particle numbers N and rod diameters a , and calculate the observables i.e. energy, density profile, and cloud size;
6. implement LDA using the hard-rod equation of state, which accounts for excluded volume;
7. compare microscopic Monte Carlo results with LDA predictions for the mentioned observables; and
8. analyze statistical uncertainties, autocorrelation effects, and convergence properties of the Monte Carlo simulations.

1.2 Requirements and Specifications

The physical and numerical requirements were:

- use harmonic oscillator units, so that a system is specified by the number of particles N and the diameter of the rod a in units of the harmonic oscillator length a_{ho} ;
- cover two values of the number of particles $N = 10$ and 20 at characteristic values of the interaction parameter $a/a_{\text{ho}} = 0, 0.1, 1$, and 10 ;
- calculate the ground-state energy, one-body density, RMS radius, and compare them with LDA based on the hard-rod equation of state;
- check the VMC sampling and kinetic energy estimators before using the trial wave function in DMC calculations;
- compare the Monte Carlo calculations with the exact hard-rod results and a finite-difference reference for two trapped rods; and
- estimate statistical errors and examine the dependence on time step, walker population, and forward time.

The code was written in Python with NumPy, SciPy, Numba, and Matplotlib. The calculations were run on a ten-core Apple M4 processor with 24 GB of memory. Each DMC result in the final set was obtained by using five independent random seeds.

1.3 Methods and Procedures

The hard-rod Hamiltonian was first written in harmonic oscillator units. The homogeneous equation of state was then used to calculate the LDA density, energy, and RMS radius. Exact hard-rod results and a finite-difference calculation for two trapped rods provided references for the Monte Carlo calculations.

VMC method was implemented first because it tests the trial wave function directly. Random-walk Metropolis (RWM) and the Metropolis-adjusted Langevin algorithm (MALA) were used to sample the same $|\Psi_T|^2$ distribution and to compare two kinetic energy estimators. The same trial wave function was then used for importance sampling in DMC, where a Metropolis-corrected Gaussian plus drift move evolves a fixed number of walkers. Forward walking gives the density and mean square radius.

After the exact cases had been checked, the calculation was run for the eight trapped systems. Selected cases were repeated with a different time step, walker population, or forward time. These additional runs show whether the differences from LDA are larger than the numerical changes produced by the calculation settings.

1.4 Gantt Diagram

Figure 1.1 summarizes the work from May to August 2026. The tasks overlapped because results from a numerical check sometimes required a change to the implementation or to the calculation settings.

The model, literature review, and the first implementation of VMC method were completed in May. DMC method, LDA, and the exact reference checks followed during May and June. Most of the eight-system calculations were carried out in June and July, while the sensitivity calculations and final VMC comparison continued into August. Writing began in July and continued alongside the last numerical checks.

Task	May	June	July	August
Literature and model	■	■		
VMC and DMC implementation	■	■	■	
LDA and exact checks	■	■	■	
Eight-system DMC calculation		■	■	
Numerical analysis			■	■
VMC validation				■
Thesis writing			■	■

Figure 1.1: Project schedule. Colored cells show the months in which each task was conducted.

1.5 Deviations and Incidences

The main objectives did not change during the project. The comparison was limited to the eight combinations of $N = 10, 20$ and $a/a_{ho} = 0, 0.1, 1, 10$. Instead of adding more particle numbers, selected cases were repeated at different time steps, walker populations, and forward times. This gave a smaller set of results with better numerical checks.

The largest rod diameter required smaller time steps and, in some cases, more walkers. For $N = 20, a/a_{ho} = 10$, the available calculations do not support an energy extrapolated to zero time step. Forward walking also becomes noisier at long forward times. The density and RMS radius are therefore reported over the finite ranges tested, and these limits are stated with the results.

2 State of the Art

Only three parts of earlier work are needed here: the exact equation of state of homogeneous hard rods, its use in LDA, and quantum Monte Carlo methods for ground-state calculations.

2.1 Homogeneous Hard Rods

For homogeneous hard rods, the ground-state energy is known exactly. On a ring of length L , the rods occupy a total length Na and leave $L' = L - Na$ for their motion; a change to rod coordinates then relates the problem to an ideal Fermi gas in this remaining length [1, 2]. The resulting hard-rod equation of state is used in the LDA calculation below.

2.2 Hard Rods in a Harmonic Trap

The harmonic trap makes the density nonuniform, so generally, there is no analytic solution in the many-body case. Still, two reference cases are available. In the hard point limit, $a = 0$, Bose–Fermi mapping gives the hard-point energy and density [3, 1, 4] which are the same as for ideal fermions. Also, for two particles, $N = 2$ at nonzero rod diameter, the Hamiltonian separates into center of mass and relative motion [5]. The relative equation is solved numerically below and used with the hard point results to check the DMC calculation.

Previous studies applied LDA to trapped one-dimensional systems by using the equation of state of the corresponding homogeneous system [6, 7, 8]. For hard rods, LDA gives the energy, density profile, and cloud size while accounting for the space occupied by the rods. Its density profile is smooth and cannot reproduce the N maxima of a finite trapped system.

2.3 Quantum Monte Carlo Methods

The microscopic calculation uses VMC and DMC techniques [9]. VMC approach samples the trial wave function and is used to compare the two kinetic energy estimators, while the importance-sampled DMC method gives the ground-state energy. Forward walking follows walker descendants to obtain the density and mean square radius with less dependence on the trial wave function [10, 11].

3 Theoretical Background

The Hamiltonian, units, observables, exact reference values, and LDA equations used in the calculations are set out below.

3.1 Hard-Rod Hamiltonian

The system contains N identical bosons of mass m in a one-dimensional harmonic trap of frequency ω . Every rod has diameter a , and the Hamiltonian is

$$H = -\frac{\hbar^2}{2m} \sum_{i=1}^N \frac{\partial^2}{\partial x_i^2} + \frac{m\omega^2}{2} \sum_{i=1}^N x_i^2 + \sum_{i<j} V_{\text{HR}}(|x_i - x_j|), \quad (3.1)$$

with

$$V_{\text{HR}}(r) = \begin{cases} +\infty, & r < a, \\ 0, & r \geq a. \end{cases} \quad (3.2)$$

Since hard rods cannot pass one another in one dimension, their order is preserved and the calculation can be restricted to

$$x_1 < x_2 < \cdots < x_N, \quad x_{i+1} - x_i \geq a. \quad (3.3)$$

The wave function vanishes at contact. In the Monte Carlo calculation, any move that crosses the boundary in Eq. (3.3) is rejected. The particle labels therefore keep their order throughout a run.

Throughout this thesis, $a = 0$ denotes the hard point limit $a \rightarrow 0^+$ with the contact condition retained. It does not denote noninteracting bosons.

3.2 Harmonic Oscillator Units

The calculation uses the natural length and energy scales

$$a_{\text{ho}} = \sqrt{\frac{\hbar}{m\omega}}, \quad E_{\text{ho}} = \hbar\omega. \quad (3.4)$$

Introduce $q_i = x_i/a_{\text{ho}}$ and divide Eq. (3.1) by $\hbar\omega$. The Hamiltonian then becomes

$$\frac{H}{\hbar\omega} = -\frac{1}{2} \sum_{i=1}^N \frac{\partial^2}{\partial q_i^2} + \frac{1}{2} \sum_{i=1}^N q_i^2, \quad q_{i+1} - q_i \geq \frac{a}{a_{\text{ho}}}. \quad (3.5)$$

These units are used throughout. Length is measured in a_{ho} , energy in $\hbar\omega$, and imaginary time in $1/\omega$, so the pair $(N, a/a_{\text{ho}})$ is enough to specify a system.

3.3 Observables

The one-body density profile of the trapped system is

$$n(x) = \left\langle \sum_{i=1}^N \delta(x - x_i) \right\rangle, \quad \int_{-\infty}^{\infty} n(x) dx = N. \quad (3.6)$$

The normalization is fixed to the number of particles N .

The cloud size is measured by the root mean square (RMS) radius,

$$R_{\text{rms}} = \left[\frac{1}{N} \left\langle \sum_{i=1}^N x_i^2 \right\rangle \right]^{1/2}. \quad (3.7)$$

In oscillator units, the density profile becomes $\tilde{n}(q) = a_{\text{ho}} n(a_{\text{ho}} q)$ and its integral also gives the total number of particles N .

Also, the free space between neighboring rods is measured,

$$g_i = q_{i+1} - q_i - \frac{a}{a_{\text{ho}}} \geq 0 \quad (3.8)$$

This distance is called the free gap. Its trap-weighted mean is

$$\bar{g}_w = \frac{\sum_{i=1}^{N-1} i(N-i)g_i}{\sum_{i=1}^{N-1} i(N-i)}. \quad (3.9)$$

The weight $i(N-i)$ is largest near the center of the trap. Both $\bar{g}_w$ and the distribution of free gaps are used to compare the two VMC samplers.

3.4 Hard Point and Two-Body References

The first DMC reference is the hard point limit. Bose–Fermi mapping gives its exact trapped ground state, and the density is the sum over the lowest N harmonic oscillator orbitals,

$$n_{\text{TG}}(x) = \sum_{k=0}^{N-1} |\varphi_k(x)|^2, \quad (3.10)$$

while the energy and RMS radius are

$$\frac{E_{\text{TG}}}{\hbar\omega} = \frac{N^2}{2}, \quad \frac{R_{\text{rms,TG}}}{a_{\text{ho}}} = \sqrt{\frac{N}{2}}. \quad (3.11)$$

These expressions provide exact references for the DMC energy, density, and RMS radius.

The second reference checks DMC results at nonzero rod diameter. For $N = 2$, define $Q = (q_1 + q_2)/2$ and $r = q_2 - q_1$. The center of mass and relative motion separate, and Eq. (3.5) becomes

$$\frac{H_Q}{\hbar\omega} = -\frac{1}{4} \frac{d^2}{dQ^2} + Q^2, \quad \frac{H_r}{\hbar\omega} = -\frac{d^2}{dr^2} + \frac{r^2}{4}, \quad (3.12)$$

The center of mass contributes $E_Q = \hbar\omega/2$. The relative equation in Eq. (3.12) is solved on a finite-difference grid. The relative wave function is set to zero at $r = a/a_{\text{ho}}$ and at the far end of the grid. Diagonalizing the resulting Hamiltonian gives the relative ground-state energy,

to which the center of mass contribution is added. Section 5.1.2 compares this numerical reference with DMC results, and Appendix B.1 checks its grid spacing.

3.5 Homogeneous Hard Rods and the LDA

For a homogeneous system with linear density n , the exact hard-rod ground-state energy per particle is [2]

$$\epsilon_{\text{HR}}(n) = \frac{\pi^2 \hbar^2 n^2}{6m(1 - an)^2}, \quad 0 \leq an < 1. \quad (3.13)$$

The factor $(1 - an)^{-2}$ accounts for the length occupied by the rods. Differentiating the energy density gives the chemical potential

$$\mu_{\text{HR}}(n) = \frac{\partial[n\epsilon_{\text{HR}}(n)]}{\partial n} = \frac{\pi^2 \hbar^2 n^2 (3 - an)}{6m(1 - an)^3}. \quad (3.14)$$

LDA applies the homogeneous equation of state at each point in the trap. The local chemical potential and the harmonic potential then add to the same constant across the cloud,

$$\mu_{\text{HR}}(n_{\text{LDA}}(x)) + \frac{1}{2}m\omega^2 x^2 = \mu_0, \quad \int n_{\text{LDA}}(x) dx = N, \quad (3.15)$$

The normalization fixes μ_0 , after which Eq. (3.15) gives the density profile. The corresponding energy and RMS radius are

$$E_{\text{LDA}} = \int \left[n_{\text{LDA}}(x) \epsilon_{\text{HR}}(n_{\text{LDA}}(x)) + \frac{1}{2}m\omega^2 x^2 n_{\text{LDA}}(x) \right] dx, \quad (3.16)$$

$$R_{\text{rms,LDA}} = \left[\frac{1}{N} \int x^2 n_{\text{LDA}}(x) dx \right]^{1/2}. \quad (3.17)$$

The LDA equations are solved numerically for every system. The grid refinement is reported in Appendix C.2.

4 Methodology

4.1 Trial Wave Function

Before writing the trial wave function, the length occupied by the rods is removed with coordinates that leave the center of the system unchanged:

$$y_i = q_i - \frac{a}{a_{\text{ho}}} \left(i - \frac{N+1}{2} \right), \quad Y = \frac{1}{N} \sum_{i=1}^N y_i, \quad u_i = y_i - Y. \quad (4.1)$$

The shift in y_i subtracts the rod lengths to the left of particle i and, because it is symmetric about the origin, leaves the mean Y unchanged. The coordinates u_i describe motion relative to that mean.

The trial wave function used in the calculation is

$$\Psi_{\text{T}}(\mathbf{q}; \alpha) \propto \exp \left[-\frac{N}{2} Y^2 - \frac{\alpha}{2} \sum_{i=1}^N u_i^2 \right] \prod_{i < j} |y_j - y_i|. \quad (4.2)$$

The pair product makes the wave function vanish at contact. The first Gaussian is the exact center of mass part, while α controls the width of the relative coordinates. When $a = 0$ and $\alpha = 1$, the expression becomes the exact hard point wave function. The logarithmic derivatives needed by VMC and DMC methods are

$$\frac{\partial \log \Psi_{\text{T}}}{\partial q_i} = -Y - \alpha u_i + \sum_{j \neq i} \frac{1}{y_i - y_j}, \quad (4.3)$$

$$\frac{\partial^2 \log \Psi_{\text{T}}}{\partial q_i^2} = -\frac{1}{N} - \alpha \left(1 - \frac{1}{N} \right) - \sum_{j \neq i} \frac{1}{(y_i - y_j)^2}. \quad (4.4)$$

From these derivatives, the local energy is

$$\frac{E_{\text{L}}}{\hbar\omega} = \frac{1}{2} \sum_i q_i^2 - \frac{1}{2} \sum_i \left[\left(\frac{\partial \log \Psi_{\text{T}}}{\partial q_i} \right)^2 + \frac{\partial^2 \log \Psi_{\text{T}}}{\partial q_i^2} \right]. \quad (4.5)$$

The separate derivatives diverge near contact, but their singular terms cancel in the local energy. Carrying out that cancellation gives

$$\begin{aligned} \frac{E_{\text{L}}}{\hbar\omega} = & \frac{N^2}{2} + \frac{(a/a_{\text{ho}})^2 N(N^2 - 1)}{24} + \frac{a/a_{\text{ho}}}{2} \sum_{k=1}^{N-1} k(N-k) g_k \\ & + \frac{\delta_{\alpha}}{2} (N^2 - 1) - \left(\delta_{\alpha} + \frac{\delta_{\alpha}^2}{2} \right) \sum_i u_i^2, \quad \delta_{\alpha} = \alpha - 1. \end{aligned} \quad (4.6)$$

The second term depends only on the rod diameter, and the third depends on the weighted free gaps. The last two contain the variational width α . For $a = 0$ and $\alpha = 1$, all configuration-

dependent terms vanish and $E_L = N^2 \hbar \omega / 2$. At nonzero diameter, α was chosen separately for each system by minimizing the estimated variance of the local energy with correlated sampling. Independent runs using Metropolis-adjusted Langevin algorithm from compact and expanded starting configurations were then used to check the selected value. The values used in DMC calculations appear in Table 5.3.

4.2 Variational Monte Carlo Method

The VMC energy of the trial wave function is

$$E_{\text{VMC}} = \frac{\int |\Psi_T(\mathbf{q})|^2 E_L(\mathbf{q}) d\mathbf{q}}{\int |\Psi_T(\mathbf{q})|^2 d\mathbf{q}}. \quad (4.7)$$

Equation (4.7) is the average local energy over the VMC distribution. Section 5.1.1 compares results obtained with the two sampling methods introduced below.

Two different ways to generate Markov chains are used to sample $|\Psi_T|^2$. In the Random walk Metropolis (RWM) algorithm one selects one particle and proposes a displacement $\delta \sim \mathcal{U}[-s, s]$ [12, 13]. A proposal inside the allowed region is accepted with probability

$$A_{\text{RWM}}(\mathbf{q} \rightarrow \mathbf{q}') = \min \left[1, \frac{|\Psi_T(\mathbf{q}')|^2}{|\Psi_T(\mathbf{q})|^2} \right]. \quad (4.8)$$

In the Metropolis-adjusted Langevin algorithm (MALA), one moves all coordinates together with the Gaussian plus drift proposal used later in DMC calculations,

$$\mathbf{q}' = \mathbf{q} + \Delta \bar{\mathbf{v}}(\mathbf{q}) + \sqrt{\Delta} \boldsymbol{\eta}, \quad \mathbf{v} = \nabla_{\mathbf{q}} \log \Psi_T, \quad \eta_i \sim \mathcal{N}(0, 1), \quad (4.9)$$

The limited drift $\bar{\mathbf{v}}$ is defined in Eq. (4.18). Because the Gaussian proposal is not symmetric, its forward and reverse probabilities are included in the Metropolis–Hastings test.

Each VMC walker evolves as an independent Markov chain. For comparison, Metropolis-adjusted Langevin algorithm uses the same move as DMC calculations, but without weights, branching, or population resampling. It therefore samples $|\Psi_T|^2$, just like Random-walk Metropolis method. This is the DMC test without branching required by the project objectives. Both methods reject moves outside Eq. (3.3). The proposal scales are $(s, \Delta) = (1.6, 0.06)$ at $a = 0$ and $(0.8, 0.04)$ at $a/a_{\text{ho}} = 1$.

4.2.1 Kinetic Energy Estimators

The two kinetic energy estimators have the same expectation value, so their comparison checks both the derivatives of the trial wave function and the sampling near contact. The first form follows from the local energy,

$$T_{\text{loc}} = \left\langle -\frac{1}{2} \frac{\nabla^2 \Psi_T}{\Psi_T} \right\rangle = \langle E_L - V_{\text{trap}} \rangle, \quad V_{\text{trap}} = \frac{1}{2} \sum_i q_i^2. \quad (4.10)$$

A second form follows from integration by parts,

$$T_{\nabla} = \frac{1}{2} \langle |\nabla \log \Psi_T|^2 \rangle_{|\Psi_T|^2}, \quad (4.11)$$

Although $|\Psi_T|^2$ goes to zero near contact, $\nabla \log \Psi_T$ diverges there. The expectation value remains finite, but a finite sample can contain unusually large contributions to Eq. (4.11).

The calculation therefore evaluates the gradient estimator at four cutoffs,

$$T_{\nabla}(\epsilon) = \frac{1}{2} \langle |\nabla \log \Psi_T|^2 \mathbf{1}_{[g_{\min} \geq \epsilon]} \rangle, \quad g_{\min} = \min_i g_i, \quad (4.12)$$

The Markov chain itself is unchanged: configurations with $g_{\min} < \epsilon$ remain in the sample, but their contribution to $T_{\nabla}(\epsilon)$ is set to zero. A linear fit gives the value at $\epsilon = 0$. The calculation is repeated without the largest cutoff and with a quadratic term to measure the dependence on the fit.

For each seed s , define $d_s = T_{\text{loc}}^{(s)} - T_{\nabla}^{(s)}(0)$. Here $T_{\nabla}^{(s)}(0)$ is the intercept from all four cutoff values, and $s_{\text{pair}} = \text{SE}\{d_s\}$ is the standard error of the paired differences. The kinetic energy scale is

$$T_{\text{scale}} = \frac{1}{2} \left(|\overline{T_{\text{loc}}}| + |\overline{T_{\nabla}(0)}| \right). \quad (4.13)$$

This average sets a common kinetic energy scale. The five paired Random-walk Metropolis runs are considered consistent when

$$|\bar{d}| + t_{\star} s_{\text{pair}} + b_{\text{fit}} < 0.01 T_{\text{scale}}, \quad (4.14)$$

Here $t_{\star}$ is the two-sided Student critical value after accounting for the two validation systems, and b_{fit} is the largest change produced by the three cutoff fits. The 1% value is a practical limit for this validation, not an extra estimate of physical uncertainty. Statistical error and the dependence on the cutoff fit are included before that limit is applied.

Appendix A.1 reports the confidence interval for the paired difference and, separately, the spread among the three cutoff fits.

4.2.2 VMC Calculation Parameters

The VMC checks are performed for two systems with $N = 10$. The trial wave function is exact in the hard-point case, while $a/a_{\text{ho}} = 1$ tests the calculation at nonzero rod diameter. Each sampler uses five independent seeds. Within a seed, 512 walkers evolve independently. The first 5,000 transitions are discarded, and the next 40,000 give 2,000 block means per seed.

Rank-normalized and folded $\hat{R}$, effective sample sizes, and Monte Carlo errors corrected for autocorrelation are calculated for every chain. The comparison includes five scalar quantities, the density, and the free gap distribution. Appendix A gives the numerical details.

4.3 Diffusion Monte Carlo Method with Importance Sampling

4.3.1 Projection in Imaginary Time

DMC calculations evolve the trial state in imaginary time τ , during which the excited-state components decay faster than the ground-state component,

$$\Phi(\mathbf{q}, \tau) = e^{-\tau(H-E_T)/\hbar} \Phi(\mathbf{q}, 0) \xrightarrow{\tau \rightarrow \infty} c_0 e^{-\tau(E_0-E_T)/\hbar} \Phi_0(\mathbf{q}). \quad (4.15)$$

Here c_0 is the overlap of the initial state with the ground state and must be nonzero. The reference energy E_T changes the overall normalization during the projection but not the ground state obtained at long imaginary time.

Importance sampling propagates $f(\mathbf{q}, \tau) = \Psi_T(\mathbf{q})\Phi(\mathbf{q}, \tau)$ rather than Φ alone. Here ω is the angular frequency of the harmonic trap, so $\tau\omega$ is dimensionless. In the ordered sector,

$$-\frac{\partial f}{\partial(\tau\omega)} = -\frac{1}{2}\nabla^2 f + \nabla \cdot (\mathbf{v}f) + \left(\frac{E_L - E_T}{\hbar\omega} \right) f, \quad \mathbf{v} = \nabla \log \Psi_T. \quad (4.16)$$

Equation (4.16) contains diffusion, drift, and a branching weight set by the local energy. The trial wave function guides the walkers toward important regions of configuration space but, in the limits of long projection time, zero time step, and an infinite walker population, does not change the projected ground state.

4.3.2 Gaussian Plus Drift Move

At each dimensionless time step $\Delta\tau\omega$, all coordinates of a walker are moved together according to

$$\mathbf{q}' = \mathbf{q} + \Delta\tau\omega \bar{\mathbf{v}}(\mathbf{q}) + \sqrt{\Delta\tau\omega} \boldsymbol{\eta}, \quad \eta_i \sim \mathcal{N}(0, 1). \quad (4.17)$$

The bare drift diverges near contact. At nonzero rod diameter it is replaced component by component by

$$\bar{v}_i = \frac{2v_i}{1 + \sqrt{1 + 2(\Delta\tau\omega)v_i^2}}, \quad (4.18)$$

The modified drift in Eq. (4.18) keeps the drift displacement finite [14]. The Metropolis–Hastings step uses the short-time Green function in both directions. A move across the hard-core boundary is rejected, leaving the position and local energy unchanged. The drift limit changes the proposal, but not the Hamiltonian or the zero time step limit.

4.3.3 Branching and Fixed Population

After the drift and diffusion move, the symmetric short-time branching factor is

$$w(\mathbf{q} \rightarrow \mathbf{q}') = \exp \left\{ -\Delta\tau\omega \left[\frac{E_L(\mathbf{q}) + E_L(\mathbf{q}')}{2\hbar\omega} - \frac{E_T}{\hbar\omega} \right] \right\}. \quad (4.19)$$

At each step, E_T is set to the population average of the local energy. It cancels when the weights are normalized. For an accepted move, Eq. (4.19) uses the old and new local energies. For a rejected move, it uses the old value twice. The calculation is repeated at several time steps to measure the remaining time-step error.

The implementation used in this thesis keeps a fixed number M of walkers throughout the calculation [15]. The walker weights change between resampling events and when the effective sample size of the weights falls below $0.35M$, systematic resampling produces M equally weighted walkers. Selected calculations are repeated with $2M$ walkers to measure the effect of the finite population.

The mixed estimator of the energy is

$$E_{\text{DMC}} = \frac{1}{T} \sum_{t=1}^T \sum_{k=1}^M \pi_{tk} E_L(\mathbf{q}_{tk}), \quad \sum_{k=1}^M \pi_{tk} = 1, \quad (4.20)$$

Here T is the number of measurements, t labels one measurement, and π_{tk} is the normalized weight of walker k . The time step and walker population are examined separately because both can shift this estimate.

4.4 Forward Walking

The mixed estimator is suitable for the energy, but estimates of the density and R^2 still depend on the trial wave function. Forward walking reduces that dependence by following each walker's descendants for a time ℓ ; several positive forward times are averaged within each run, and the five independent runs are combined afterward.

Each walker records its parent. The implementation carries a measured quantity through the successive parent maps until the selected forward time has elapsed:

$$\begin{aligned} A_k^{(s;s)} &= O(\mathbf{q}_k^{(s)}), \\ A_k^{(s;u+1)} &= A_{p_k^{(u+1)}}^{(s;u)}, \quad u = s, \dots, s+L-1, \\ \hat{O}_{s,\ell} &= \sum_{k=1}^M \pi_k^{(s+L)} A_k^{(s;s+L)}. \end{aligned} \quad (4.21)$$

The measurement is made at step s , and $L = \ell/\Delta\tau$ is the number of steps in the forward time. Whenever resampling occurs, the parent index $p_k^{(u+1)}$ carries the stored value to the descendant walker. Between resampling events, a walker is its own parent. After L steps, the transported values are averaged with the endpoint weights $\pi_k^{(s+L)}$. For R^2 , the stored quantity is the relative contribution from one walker; the exact center of mass part is added later. For the density, the stored coordinates follow the same parent maps and are placed in the histogram after the final map.

As the forward time grows, resampling leaves fewer independent families. The calculation records both the effective number of families that still have descendants and the fraction of weight carried by the largest family. A result is retained only if the first value remains above 50 and the second below 10%, including their uncertainties. These conditions support the finite forward times used here, but they do not prove convergence as $\ell \rightarrow \infty$; Appendix C lists the selected times.

4.5 Exact Center of Mass Averaging

For a harmonic trap, the center of mass separates exactly from the relative coordinates. With $Q = N^{-1} \sum_i q_i$,

$$\frac{R_{\text{rms}}^2}{a_{\text{ho}}^2} = \left\langle \frac{1}{N} \sum_i (q_i - Q)^2 \right\rangle + \frac{1}{2N}, \quad \langle Q^2 \rangle = \frac{1}{2N}. \quad (4.22)$$

The center of mass contribution to R^2 is known exactly. Forward walking is therefore applied only to the relative part, and $1/(2N)$ is added afterward. For the density, the same separation gives

$$n(q) = \int_{-\infty}^{\infty} \sqrt{\frac{N}{\pi}} e^{-NQ^2} n_{\text{rel}}(q - Q) dQ, \quad (4.23)$$

Here n_{rel} is formed after subtracting the center of mass of each walker, and the Gaussian has variance $1/(2N)$. The density also satisfies $n(x) = n(-x)$, so the two sides of the trap are averaged. Both operations reduce noise without changing the relative motion sampled by DMC approach.

5 Results

First, the variational results are checked for internal consistency and are tested against known results. Second, the DMC results are validated. This is followed by the eight systems, their numerical uncertainties, and the comparison with LDA.

5.1 Validation and Calculation Settings

5.1.1 VMC Validation

Table 5.1 shows the VMC results obtained using two different sampling methods. In Random-walk Metropolis method, only one particle is moved at a time with a uniform random step, whereas in Metropolis-adjusted Langevin algorithm all particles are moved together using a Gaussian step guided by the drift. The Random-walk Metropolis acceptance rule and the Metropolis-adjusted Langevin algorithm proposal are given in Eqs. (4.8) and (4.9), respectively. The two samplers give almost identical total energies E and the mean squared cloud size R^2 . Their kinetic and trap energies are not compared by raw standard errors alone; the practical comparison in Eq. (4.14) also includes the cutoff fit and its uncertainty. At nonzero rod diameter, the difference between the two VMC energies is much smaller than the change from the VMC method to the DMC method.

For zero hard rod diameter, $a = 0$, the trial wave function is exact, so the local energy is constant and both samplers return the exact value $50\hbar\omega$. At $a/a_{\text{ho}} = 1$, the VMC energy is about $0.0106\hbar\omega$ above the DMC result in Table 5.4.

For the two kinetic estimators, the left-hand side of Eq. (4.14) is $0.1161\hbar\omega$, below $0.2498\hbar\omega$, at $a = 0$, and $0.3829\hbar\omega$, below $0.4041\hbar\omega$, at $a/a_{\text{ho}} = 1$. The second comparison is only narrowly within the 1% criterion. Appendix A.1 gives the cutoff fits and the dependence on the choice of fit.

Figure 5.1 shows the density profile. At $a = 0$, the relative L^2 errors against the exact density are 0.00548 for Random-walk Metropolis method and 0.00111 for Metropolis-adjusted Langevin algorithm. Both samplers reproduce its oscillations and give ten maxima at $a/a_{\text{ho}} = 1$.

The largest rank-normalized and folded $\hat{R}$ is 1.00224, total bulk effective sample size (ESS) is at least 1380, and the smallest ESS value for one seed is 205. One Random-walk Metropolis

Table 5.1: VMC approach estimates from five independent RWM and MALA calculations. Energies are in $\hbar\omega$, and R^2 is in a_{ho}^2 . Parentheses give one standard error over the five runs.

N	a/a_{ho}	sampler	E_{VMC}	T_{loc}	V_{trap}	R^2/a_{ho}^2
10	0	RWM	50 (exact)	24.99210(27)	25.00790(27)	5.00158(55)
10	0	MALA	50 (exact)	24.99943(23)	25.00057(23)	5.00011(47)
10	1	RWM	146.84790(24)	40.40981(32)	106.43808(32)	21.28762(64)
10	1	MALA	146.847841(59)	40.40654(25)	106.44130(25)	21.28826(50)

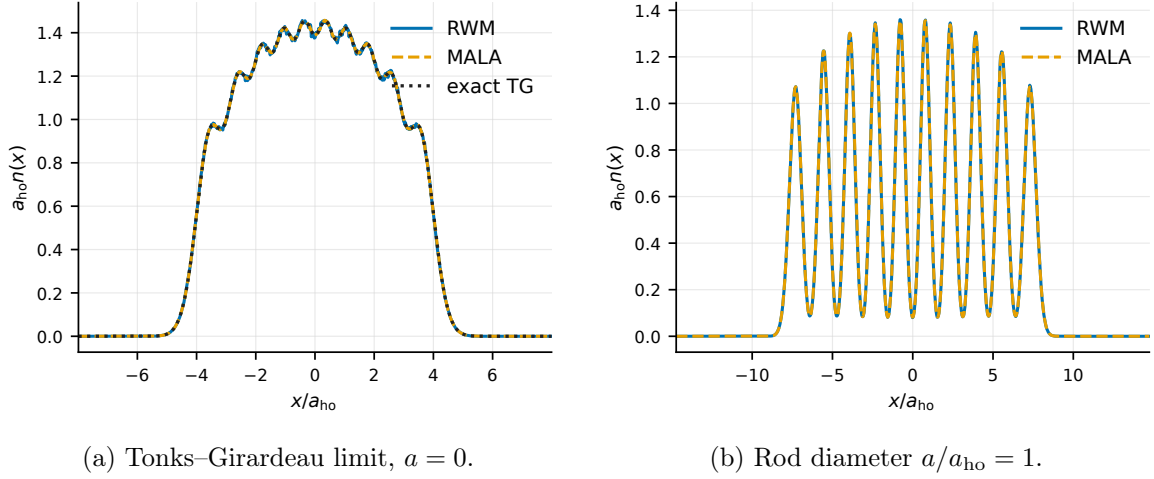


Figure 5.1: VMC densities from RWM (blue) and MALA (orange), averaged over five independent runs. The dotted curve in panel (a) is the exact TG density. The statistical error is smaller than the line width.

Table 5.2: VMC sampler and convergence measures. The kinetic column compares the value from Eq. (4.14) with its 1% criterion. Reported ESS values are minima over the traces and $\hat{R}$ is the maximum.

a/a_{ho}	p_{acc}^{RWM}	p_{acc}^{MALA}	kinetic value/criterion	$\max \hat{R}$	min ESS (total/seed)
0	0.2766	0.4882	0.1161/0.2498	1.00224	1399/232
1	0.4187	0.4552	0.3829/0.4041	1.00145	1380/205

free gap trace at $a = 0$ shows a weak trend, but it does not change the result obtained from the five runs. Table 5.2 collects the acceptance rates and the most restrictive convergence values; together, they show that both samplers explored their distributions adequately. Appendix A gives the remaining comparisons.

The agreement between the samplers checks that both implementations sample $|\Psi_T|^2$. It does not imply that the trial wave function is the exact ground state at nonzero rod diameter.

5.1.2 DMC Validation with Exact References

At $a = 0$, the mixed energies are exactly $50\hbar\omega$ for $N = 10$ and $200\hbar\omega$ for $N = 20$. Forward walking gives radii of $2.23614(19)a_{ho}$ and $3.16215(18)a_{ho}$, in agreement with Eq. (3.11). After averaging the exact density over the same bins, the relative L^2 errors are 9.28×10^{-4} and 9.41×10^{-4} . The hard point calculation therefore checks the energy, density, and radius together.

For $N = 2$, the finite-difference calculation with 2,000 points gives $2.999987\hbar\omega$ at $a/a_{ho} = 1$ and $32.577211\hbar\omega$ at $a/a_{ho} = 10$. The corresponding DMC values are $2.9857(89)\hbar\omega$ and $32.5788(76)\hbar\omega$. Both lie within two Monte Carlo standard errors of the numerical reference. Appendix B.1 shows that the grid error is smaller than either standard error.

5.1.3 DMC Calculation Settings

The main set consists of the eight combinations $N = 10, 20$ and $a/a_{\text{ho}} = 0, 0.1, 1, 10$, each calculated with five independent seeds. A run discards $\tau_{\text{heat}}\omega = 60$ and measures the following $\tau_{\text{prod}}\omega = 480$ at intervals of $0.025/\omega$. Other settings are in Table 5.3. Wider density histograms are used as the rods grow, and none of the reported profiles reaches a histogram boundary.

Table 5.3: Settings used in the eight DMC calculations. The exact TG trial wave function has no fitted width. The extra digits in α record the values used in the calculations, not their physical precision.

N	a/a_{ho}	α	$\Delta\tau\omega$	M	steps between measurements	q_{max}	bins
10	0	not applicable	0.002500	256	10	35.000	840
10	0.1	1.063733	0.002500	256	10	35.500	840
10	1	1.622444	0.001250	512	20	40.000	840
10	10	5.590865	0.000125	256	200	178.871	3579
20	0	not applicable	0.002500	256	10	35.000	840
20	0.1	1.090809	0.002500	256	10	36.000	840
20	1	1.866936	0.000625	512	40	48.512	972
20	10	7.011111	0.000250	512	100	353.565	7073

In Table 5.3, q_{max} is the positive edge of the symmetric density grid and “bins” gives its resolution. The selected time steps give acceptance rates between 93.2% and 99.7%. Such high acceptance means that the proposed moves are small, not that the calculation is automatically accurate. With larger time steps, acceptance falls to 42.4%–91.1% and the walkers move farther in each step. A larger time step also changes the short-time approximation, so acceptance alone cannot select the final value. The time-step comparisons below are used for that purpose. Appendix B.2 gives the larger-step results.

5.1.4 Statistical Analysis and Numerical Effects

Every reported Monte Carlo value is a mean over five independent runs. Three error estimates are calculated for the energy: the spread over runs, the blocking result, and an estimate based on the integrated autocorrelation time [16, 17]. The largest is reported. Split $\hat{R}$ and effective sample size are checked separately [18].

The early and late parts of each run were also compared, and two seeds for $N = 10, a/a_{\text{ho}} = 0.1$ show a trend. When the eight systems are tested together, the largest standardized change is 3.015, below the simultaneous limit of 3.529. For that system, split $\hat{R}$ is 1.00075 and the smallest ESS for one seed is 1030; the lowest ESS anywhere in the final set is 944. The DMC value of split $\hat{R}$ was calculated without rank normalization; the VMC diagnostic above was rank normalized and folded.

The numerical study varies the time step, doubles the walker population, and checks whether the population change depends on the time step. The density and RMS radius are also compared at different forward times and at selected alternative time steps and populations.

Table 5.4: Selected DMC and LDA energies and RMS radii. Energies are in $\hbar\omega$, radii are in a_{ho} , and differences are in percent. Five finite-diameter energies are extrapolated to zero time step; the $N = 20, a/a_{\text{ho}} = 10$ value remains at finite time step.

N	a/a_{ho}	Energy			RMS radius		
		E_{DMC}	E_{LDA}	δ_E	$R_{\text{rms,FW}}$	$R_{\text{rms,LDA}}$	δ_R
10	0	50	50	0.000	2.23614(19)	2.23607	0.003
10	0.1	56.643	56.760	-0.206	2.45164(16)	2.45547	-0.156
10	1	146.837	148.227	-0.938	4.61475(9)	4.63609	-0.460
10	10	4535.877	4583.431	-1.038	29.65600(13)	29.80971	-0.516
20	0	200	200	0.000	3.16215(18)	3.16228	-0.004
20	0.1	238.806	238.993	-0.078	3.60206(28)	3.60325	-0.033
20	1	838.072	840.377	-0.274	8.07237(15)	8.08486	-0.155
20	10	35333.614	35425.345	-0.259	58.85079(10) [†]	58.92667	-0.129

[†] The $N = 20, a/a_{\text{ho}} = 10$ RMS radius comes from a second calculation with identical physical and numerical settings and the same random seeds. Only R^2 is taken from this calculation.

Appendix B.3 defines the four separate 95% bounds used for the energy. A comparison at two populations is treated as a sensitivity check, not as an extrapolation to $M \rightarrow \infty$.

5.2 Energy and RMS Radius

Define the relative differences

$$\delta_E = 100 \left(\frac{E_{\text{DMC}}}{E_{\text{LDA}}} - 1 \right), \quad \delta_R = 100 \left(\frac{R_{\text{rms,FW}}}{R_{\text{rms,LDA}}} - 1 \right). \quad (5.1)$$

Negative values of δ_E or δ_R mean that the microscopic result is below the corresponding LDA value.

Table 5.4 contains the main scalar results for all eight systems. Parentheses give one Monte Carlo standard error (MCSE), and the energy MCSE is smaller than the displayed precision in every case. Extrapolated energies are rounded to the resolution supported by the numerical analysis. The energy fits and forward walking calculations are described in Appendices B.3 and C.

All six DMC energies at $a > 0$ lie below the LDA values. The largest relative difference is 1.038% for $N = 10, a/a_{\text{ho}} = 10$. In every case, the difference is larger than both the Monte Carlo standard error and the four numerical bounds. The forward walking radii are also smaller in their central values, with a largest difference of 0.516%.

One radius comparison is too close to call. For $N = 20, a/a_{\text{ho}} = 0.1$, the forward walking radius is $0.001199a_{\text{ho}}$ below LDA, but changing the numerical settings moves it by as much as $0.001469a_{\text{ho}}$. The calculation therefore does not establish the sign of this small difference.

At fixed diameter, agreement with LDA improves from $N = 10$ to $N = 20$. At $a/a_{\text{ho}} = 1$,

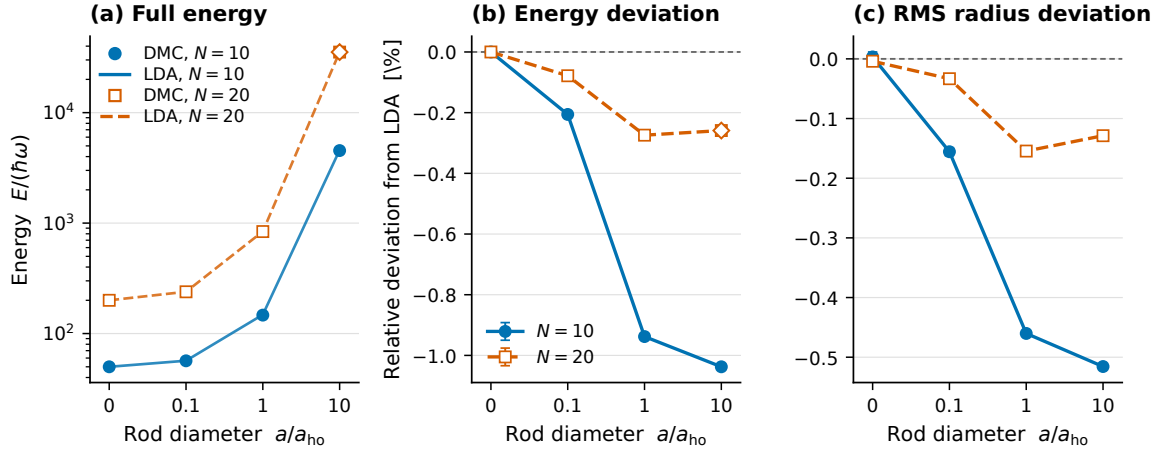


Figure 5.2: Energy and RMS radius for the eight systems. Panel (a) shows the DMC and LDA energies. Panels (b) and (c) show their relative differences. The open diamond is the energy at the selected finite time step for $N = 20, a/a_{ho} = 10$, where no zero-step intercept is reported. Error bars are smaller than the symbols.

for example, the magnitude of the energy difference falls from 0.938% to 0.274%, and the radius difference falls from 0.460% to 0.155%. Two particle numbers are not enough to infer a scaling law, but the trend is the same at the three nonzero diameters.

The energy scale changes strongly with rod diameter. Between $a/a_{ho} = 0$ and 10, the energy rises by almost two orders of magnitude for $N = 10$ and by more than two for $N = 20$. The density profiles below also show the corresponding increase in the size of the cloud.

5.3 Density Profiles and Finite-N Shell Structure

Figure 5.3 shows the density profile as obtained using Local Density Approximation (LDA) and by using the forward-walking technique, averaged over $\ell\omega = 2, 4, 7$. Every curve integrates to N . Center of mass and parity averaging are included, and the blue bands show one pointwise standard error over five runs.

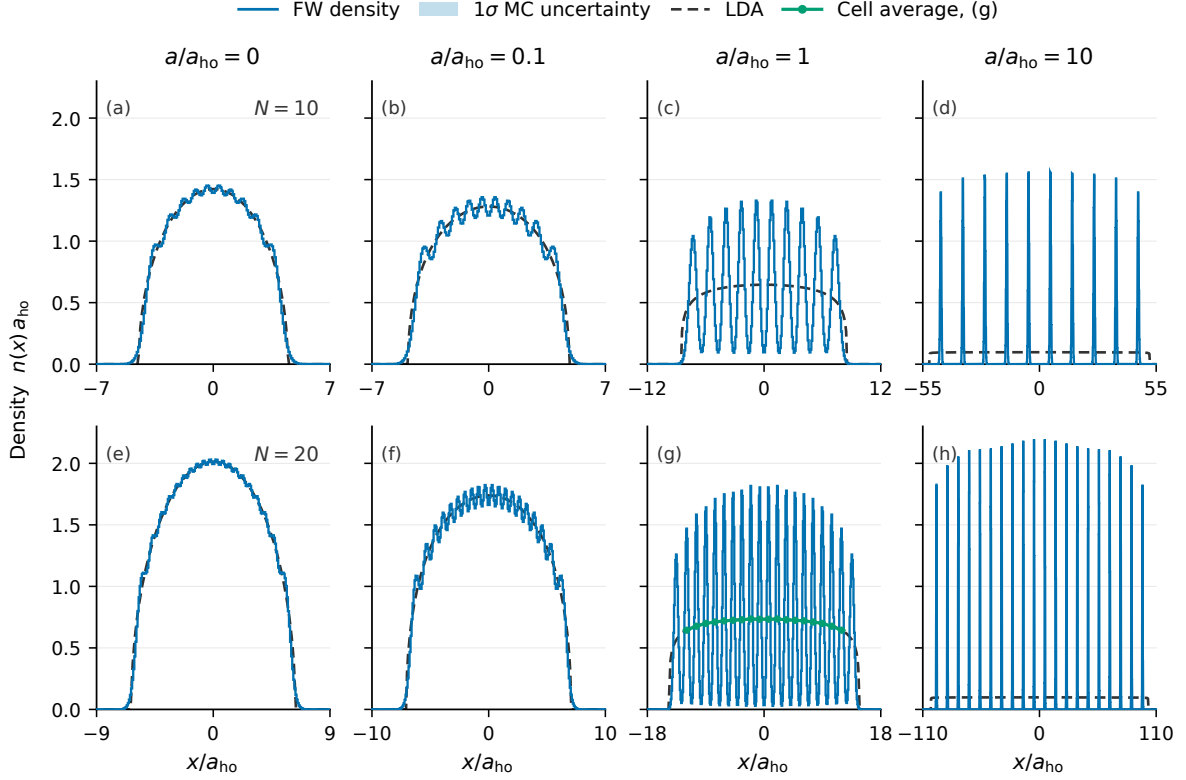


Figure 5.3: Forward walking densities (blue) and LDA profiles (black dashed) for the eight systems. The blue bands show one pointwise standard error. In panel (g), the green curve is the average over cells containing one particle, defined in Eq. (5.3).

At $a = 0$, the relative L^2 error against the exact finite N density is below 10^{-3} , while LDA gives a smooth curve and misses the oscillations. They remain weak at $a/a_{ho} = 0.1$, where the cloud edge and central density are still similar to LDA.

At $a/a_{ho} = 1$, each density profile has N clear maxima; the minima between them nearly reach zero for $N = 10$ and remain deep for $N = 20$. At $a/a_{ho} = 10$, the narrow maxima are about one rod diameter apart.

Since even number of particles N are considered, the peaks occur in pairs around the center and the density has a central minimum. LDA has no such peaks. To compare the profile on a scale larger than the peak spacing, panel (g) averages the $N = 20, a/a_{ho} = 1$ density over cells that each contain one particle. The cell boundaries are defined by

$$C(x) = \int_{-\infty}^x n_{FW}(x') dx', \quad C(b_j) = j, \quad (5.2)$$

Here $C(x)$ is the expected number of particles to the left of x , so each interval contains one particle. Its mean density is

$$\bar{n}_j = \frac{1}{b_{j+1} - b_j}, \quad j = 1, \dots, N - 2. \quad (5.3)$$

Because each cell contains one particle, its mean density is the inverse of its width. The two open cells at the edges are omitted. The green line comes from the same forward walking

density as the blue curve; it is only a local average, not a second estimator. Although the original profile has narrow peaks and deep minima, the cell average follows LDA through the middle of the trap. This comparison is shown for one system. The RMS radius gives an integrated comparison for all eight.

The profile distances are listed in Appendix C.1. From the smallest to the largest nonzero diameter, the distance between forward walking and LDA grows from 0.048 to 3.19 for $N = 10$, and from 0.030 to 3.78 for $N = 20$. The forward walking and mixed densities remain much closer, with distances below 0.025. Changes produced by the tested forward times, time steps, and populations remain below 0.03.

The scalar and density comparisons describe two different length scales. LDA remains close to the energy, RMS radius, and the cell-averaged central profile, but it cannot reproduce the N maxima. The common ordering of the LDA and DMC energies is a numerical result, not a variational bound on LDA.

5.4 Numerical Uncertainties

Table 5.5 gives the energies extrapolated to zero time step at fixed walker population. The last column is the largest of the four separate 95% bounds described in Appendix B.3. The value $0.01\hbar\omega$ is used only as the resolution for deciding whether to report an intercept. It is not a replacement for the statistical error or for any of the four bounds.

Table 5.5: Energies extrapolated to zero time step at fixed walker population, in $\hbar\omega$. The last column is the largest of four separate 95% upper bounds. An intercept is shown only when all four bounds are below $0.01\hbar\omega$. They are not combined.

N	a/a_{ho}	M	$E(0, M)$	largest 95% bound
10	0.1	256	56.64324	0.00114
10	1	512	146.8372	0.0068
10	10	256	4535.8771	0.0050
20	0.1	256	238.8057	0.0027
20	1	512	838.0724	0.0043
20	10	512	not reported	0.039

Five systems with nonzero rod diameter have all four bounds below $0.01\hbar\omega$. Figure B.1 shows representative fits together with the one system for which no intercept is reported.

At $N = 20, a/a_{\text{ho}} = 10$, Table 5.4 therefore uses the energy $35333.6136(10)\hbar\omega$ at the selected finite time step. No zero-step intercept is reported. The bound from the choice of fit and the two bounds involving walker population are all larger than $0.01\hbar\omega$. Smaller time steps and at least one calculation with a larger walker population would be needed to reduce them.

The 95% relative bounds range from 3.52×10^{-6} to 4.70×10^{-4} for the radius and from 1.26×10^{-3} to 2.71×10^{-3} for the density. Every selected forward time also passes the two ancestry checks. The density and radius are stable over the settings tested, but they are not extrapolated to zero time step, infinite population, or infinite forward time.

6 Conclusions and Future Development

6.1 Conclusions

This thesis used variational Monte Carlo (VMC) and importance-sampled diffusion Monte Carlo (DMC) methods, together with the forward walking technique, to study hard rods in a one-dimensional harmonic trap. Random-walk Metropolis method and the Metropolis-adjusted Langevin algorithm give consistent VMC results, and the two kinetic energy estimators satisfy the 1% comparison. In the hard point limit, the DMC energy, density, and root mean square (RMS) radius agree with the exact Tonks–Girardeau values.

For nonzero rod diameter, the DMC energies are 0.078%–1.038% below LDA, and the radii obtained by forward walking are 0.033%–0.516% smaller. At fixed diameter, the magnitudes of both differences are smaller for $N = 20$ than for $N = 10$. The smallest radius difference is below the numerical change measured for that system, so its sign is not established.

The density shows what is missing from the scalar comparison. Its finite- N oscillations become more pronounced as the rod diameter grows, and at $a/a_{\text{ho}} = 10$ the profile contains N narrow maxima. LDA remains smooth. For $N = 20, a/a_{\text{ho}} = 1$, averaging the microscopic density over cells containing one particle recovers the LDA shape in the center of the trap. LDA therefore captures the cloud size without reproducing the finite- N shell structure.

Five systems with nonzero rod diameter have an energy extrapolated to zero time step with all four 95% bounds below $0.01\hbar\omega$. The calculation for $N = 20, a/a_{\text{ho}} = 10$ does not meet this condition, so its reported energy remains at a finite time step. The density and RMS radius change little over the settings tested, but they are not extrapolated to infinite walker population or infinite forward time.

6.2 Future Work

The first extension is a zero time step result for $N = 20, a/a_{\text{ho}} = 10$. This requires smaller time steps and at least one calculation with a larger walker population.

A wider range of forward walking lengths and a second trial wave function would test the remaining dependence of the density and RMS radius. Additional particle numbers would also show whether the improvement of LDA from $N = 10$ to 20 continues for larger clouds.

A Details of the VMC calculation

This Appendix contains the numerical details behind the VMC checks in Sec. 5.1.1. The first section gives the zero-cutoff value of the gradient kinetic estimator. The second compares the two sampling methods.

A.1 Cutoff extrapolation of the gradient kinetic energy estimator

Table A.1 gives the Random-walk Metropolis data used to reach zero cutoff. Configurations with $g_{\min} < \epsilon$ remain in the sample, but contribute zero to this estimator. Their fraction is denoted by p_{exc} . The main linear and quadratic fits use all four cutoffs. A second linear fit omits the largest one.

Table A.1: Gradient kinetic energy in $\hbar\omega$ at the four cutoffs. The last column is the fraction of configurations with $g_{\min} < \epsilon$.

a/a_{ho}	ϵ	$T_{\nabla}(\epsilon)$	p_{exc}
0	0.01	24.32826	2.30×10^{-5}
0	0.02	23.66945	1.74×10^{-4}
0	0.04	22.44080	1.31×10^{-3}
0	0.08	19.81963	1.07×10^{-2}
1	0.01	39.02267	4.35×10^{-5}
1	0.02	37.75303	3.44×10^{-4}
1	0.04	35.08692	2.79×10^{-3}
1	0.08	29.63362	2.18×10^{-2}

Linear fits through all four cutoffs give $24.97505\hbar\omega$ at $a = 0$ and $40.41739\hbar\omega$ at $a/a_{\text{ho}} = 1$. The corresponding local kinetic energies are $24.99210\hbar\omega$ and $40.40981\hbar\omega$. Table A.2 lists the statistical confidence term and the change produced by the choice of fit separately.

Table A.2: Terms in the RWM kinetic comparison, in $\hbar\omega$. The 95% confidence term includes the correction for the two cases. The fit dependence is listed separately.

a/a_{ho}	mean difference	95% confidence term	dependence on the choice of fit	left-hand side	1% criterion
0	0.01706	0.05408	0.04495	0.11608	0.24984
1	-0.00758	0.27236	0.10298	0.38291	0.40414

Both rows satisfy the 1% criterion. The comparison at $a/a_{\text{ho}} = 1$ is close to the limit, which is why the main text describes it as a narrow agreement rather than a precise equality.

A.2 Comparison of the VMC sampling methods

The convergence checks require $\hat{R} < 1.01$, total bulk effective sample size (ESS) above 400, and ESS above 50 for every seed. The Monte Carlo standard error includes autocorrelation. Practical comparison limits were set at 0.5% for total energy, 1% for kinetic energy, trap energy, and R^2 , 2% for the weighted free gap, and 5% for the two distributions. These values are validation limits rather than estimates of physical error. Profile errors are obtained by resampling the five independent runs.

Each scalar comparison uses two independent groups of five seeds. Let d be the difference between their means, h the Bonferroni-adjusted 95% confidence term, and m the comparison limit. The rule is

$$\begin{aligned} |d| + h \leq m &\Rightarrow \text{consistent within the comparison limit,} \\ |d| > m &\Rightarrow \text{different by more than the comparison limit,} \\ \text{otherwise} &\Rightarrow \text{inconclusive.} \end{aligned} \tag{A.1}$$

All scalar comparisons satisfy the first line of Eq. (A.1). For a profile, the measured relative L^2 distance is added to its bootstrap confidence term. Within each physical system, the multiple-comparison correction covers five scalar quantities and two profiles. Scalar intervals use the t distribution. Profiles are resampled by run rather than by histogram bin.

For the density, the measured relative L^2 distance and its upper bound are 0.00557 and 0.01453 at $a = 0$, and 0.00427 and 0.01085 at $a/a_{\text{ho}} = 1$. The corresponding values for the distribution of free gaps are 0.00134 and 0.00376, and 0.00109 and 0.00289. No probability mass reaches the histogram boundaries.

B DMC convergence

This appendix explains how the DMC settings were checked. It covers the two-body grid, the larger proposal steps, and the energy fits.

B.1 Two-body finite-difference convergence

The finite-difference grid runs from $r = a/a_{\text{ho}}$ to $r = a/a_{\text{ho}} + 20$, with the wave function set to zero at both ends. Halving the grid spacing reduces the change in the energy by about four, as expected for a centered second-order calculation. The values are shown in Table B.1.

Table B.1: Finite-difference energies at four grid sizes. Energies are in $\hbar\omega$; the Monte Carlo comparison uses a grid of 2,000 points.

points	Δr	$E_{a/a_{\text{ho}}=1}$	$E_{a/a_{\text{ho}}=10}$
350	0.056980	2.999586	32.574435
700	0.028531	2.999896	32.576581
1400	0.014276	2.999974	32.577119
2000	0.009995	2.999987	32.577211

The change from 1,400 to 2,000 grid points is $1.3 \times 10^{-5} \hbar\omega$ at $a/a_{\text{ho}} = 1$ and $9.2 \times 10^{-5} \hbar\omega$ at $a/a_{\text{ho}} = 10$. Both are much smaller than the corresponding statistical errors in the DMC calculations, so the grid does not limit the two-body comparison in Sec. 5.1.2.

B.2 Acceptance rate and expected squared jumping distance

The acceptance rates in the main calculations are high. This does not show how far a walker moves. Table B.2 therefore compares each setting with a larger time step and reports the expected squared jumping distance (ESJD). ESJD is the mean squared movement of the full walker configuration per attempted step, with rejected moves counted as zero.

Table B.2: Final settings (subscript 0) and larger-step comparisons (subscript c). The last column gives the ratio of the full-configuration ESJD per DMC step.

N	a/a_{ho}	$p_{\text{acc},0}$	$\Delta\tau_c\omega$	M_c	$p_{\text{acc},c}$	ESJD ratio
10	0.1	0.9838	0.040000	256	0.6238	10.40
10	1	0.9884	0.010000	512	0.8636	7.17
10	10	0.9970	0.002000	512	0.9105	14.97
20	0.1	0.9317	0.020000	256	0.4240	3.72
20	1	0.9750	0.002500	512	0.8746	3.65
20	10	0.9593	0.000625	512	0.8831	2.33

The larger steps increase ESJD by factors from 2.33 to 14.97. They can also increase the time-step error. This comparison therefore shows the cost of the small moves, while the energy fits below decide whether a time step is small enough for the reported energy.

For $N = 10, a/a_{\text{ho}} = 10$, the population comparison uses $\Delta\tau\omega = 0.001, 0.002$ and $M = 256, 512$. Energy is compared at $(0.001, 256)$. Forward walking is compared at $(0.002, 512)$ because the two quantities need different useful settings.

B.3 Energy fits and uncertainty bounds

At fixed M , the data at several time steps are fitted with

$$\begin{aligned} E_{\text{lin}}(\Delta) &= E_0 + c_1\Delta, \\ E_{\text{quad}}(\Delta) &= E_0 + c_2\Delta^2, \quad \Delta = \Delta\tau\omega. \end{aligned} \tag{B.1}$$

The linear and quadratic fits give separate estimates at zero time step. Their difference shows how much the result depends on the assumed leading power of Δ . A third form, $E_0 + c_1\Delta + c_2\Delta^2$, is used only to check for visible curvature. Its intercept is not used as a reported energy.

Each weighted fit uses the largest of the three Monte Carlo standard error estimates for an energy. If the reduced χ^2 exceeds one, the covariance is enlarged by the same factor. Different time steps use different seeds and are treated as independent. Table B.3 lists the time steps and walker population used in each fit, making the fitted range explicit.

Table B.3: Energies at fixed walker population calculated at several time steps. The final column lists DMC time steps, not forward times.

N	a/a_{ho}	M	fitted $\Delta\tau\omega$ values
10	0.1	256	0.0025, 0.005, 0.010, 0.020, 0.040
10	1	512	0.00125, 0.0025, 0.005, 0.010
10	10	256	0.000125, 0.00025, 0.0005, 0.001
20	0.1	256	0.0025, 0.005, 0.010, 0.020
20	1	512	0.000625, 0.00125, 0.002, 0.0025
20	10	512	0.00025, 0.0004, 0.0005, 0.000625

The fit is repeated without the largest time step to see whether the chosen range matters. The effect of doubling the walker population is also measured at two time steps. Table B.4 lists the four results separately. Here b_{fit} compares the linear and quadratic intercepts, b_{range} changes the fitted range, b_M compares M with $2M$, and $b_{\tau M}$ checks whether the population effect changes with the time step.

Table B.4: Uncertainty bounds for extrapolation of the energy from the mixed estimator to zero time step at nonzero rod diameter, in $\hbar\omega$. The standard error of the linear intercept is s_{E_0} . Each b is a separate 95% upper bound. No energy extrapolated to zero time step is quoted for the last row.

N	a/a_{ho}	$E(0, M)$	s_{E_0}	b_{fit}	b_{range}	b_M	$b_{\tau M}$	all four bounds below 0.01
10	0.1	56.64324	0.00004	0.00004	0.00004	0.00084	0.00114	yes
10	1	146.8372	0.0003	0.0002	0.0008	0.0060	0.0068	yes
10	10	4535.8771	0.0011	0.0020	0.0030	0.0042	0.0050	yes
20	0.1	238.8057	0.0001	0.0003	0.0004	0.0023	0.0027	yes
20	1	838.0724	0.0008	0.0024	0.0006	0.0039	0.0043	yes
20	10	35333.630	0.003	0.015	0.006	0.025	0.039	no

The last column records whether every bound is below $0.01\hbar\omega$. They are kept separate rather than added into one error bar.

For comparisons between fit forms and between fitted ranges, the displayed 95% bound is

$$b = |\hat{\theta}_1 - \hat{\theta}_2| + z_{0.975} s_{12}, \quad (\text{B.2})$$

Here $z_{0.975}$ is the 97.5th percentile of the standard normal distribution, and s_{12} is the standard error of the difference between the two fitted quantities. For the fit comparison, $\hat{\theta}_1$ and $\hat{\theta}_2$ are the linear and quadratic intercepts. The fitted-range check instead compares an intercept with the result obtained after removing the largest time step. In either case, the bound includes both the observed change and its uncertainty.

For the comparison at two walker populations, the contrast formed with matching random seeds $d_s = E_s(2M) - E_s(M)$ gives

$$b_M = |\bar{d}| + t_{0.975, \nu} \max \left[s_{\text{paired}}, \sqrt{s_M^2 + s_{2M}^2} \right], \quad \nu = n_{\text{seed}} - 1. \quad (\text{B.3})$$

The two populations must use the same ordered seed IDs. The standard error in Eq. (B.3) is the larger of the paired standard error and the quadrature sum of the two run-level conservative standard errors. The factor $t_{0.975, \nu}$ is the corresponding Student critical value. Each statement is two-sided at 95%, with no further correction across the four statements.

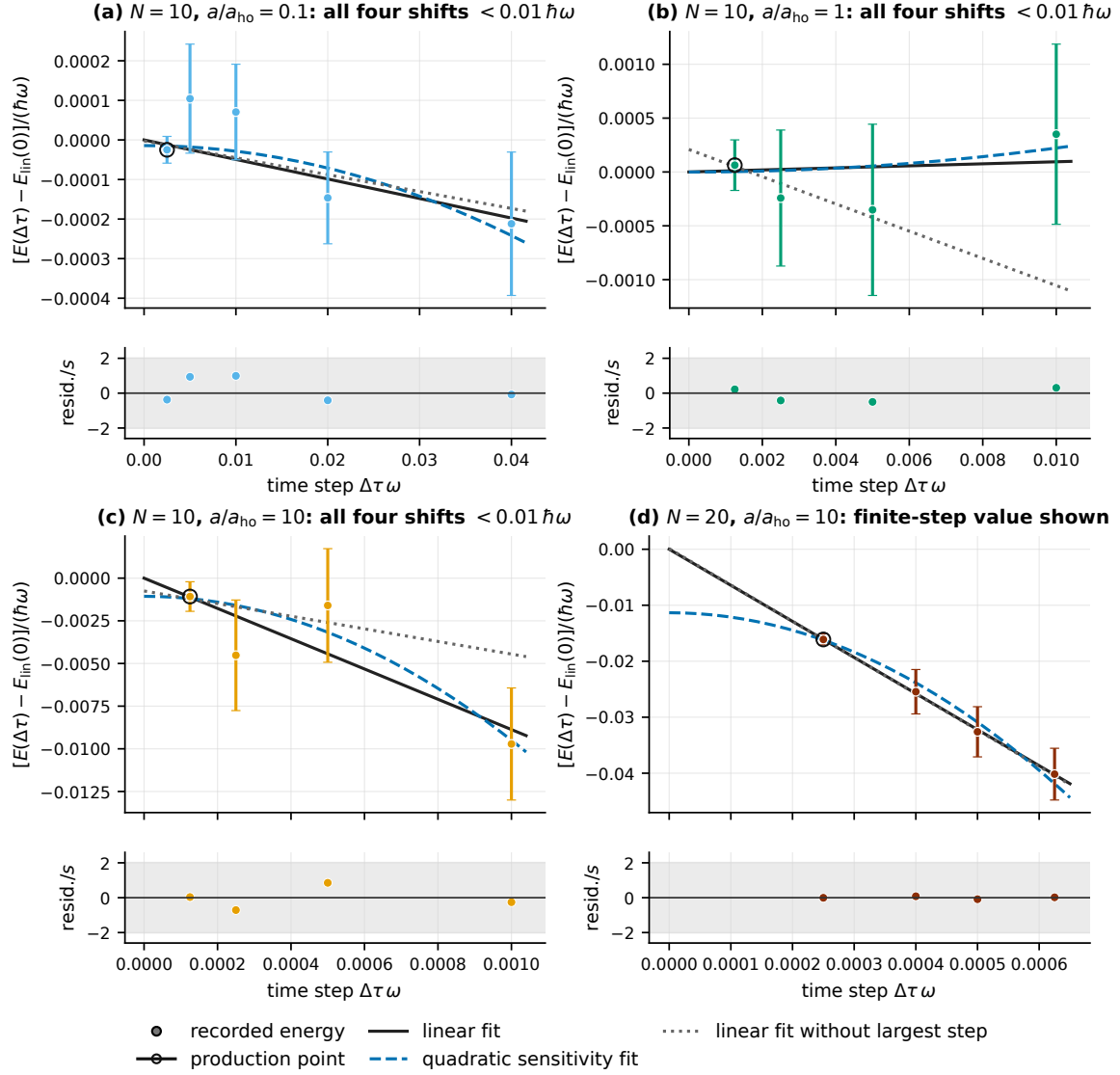


Figure B.1: Energies relative to the linear zero-step intercept. Bars show MCSE. Solid black and dashed blue lines are the linear and quadratic fits; the dotted gray line omits the largest time step. Open circles mark the settings in Table 5.3. Lower panels show the linear-fit residuals.

Panels (a)–(c) show cases for which the changes among the fitted forms remain below the reporting limit. In panel (d), the fitted lines do not support a zero-step value within that limit. This is why the $N = 20, a/a_{ho} = 10$ energy is reported only at its finite production time step.

C Density resolution and dependence on forward time

This appendix checks the spatial grid and the forward times used for the density and RMS radius.

C.1 Density distance

Profiles on the same grid are compared with

$$D(f, g) = \left[\frac{\int [f(x) - g(x)]^2 dx}{\int g(x)^2 dx} \right]^{1/2}. \quad (\text{C.1})$$

The distance is zero when the two profiles are identical. A larger value means a larger difference relative to g , and the result can be greater than one. The mixed density is included only as a comparison for forward walking. Table C.1 shows that the distance from LDA grows as the peaks narrow, while the distance from the mixed density remains small.

Table C.1: Relative L^2 distances between normalized profiles after center-of-mass and parity averaging.

N	a/a_{ho}	$D(\text{FW}, \text{LDA})$	$D(\text{FW}, \text{mixed})$
10	0	0.04843	0.00126
10	0.1	0.07408	0.00261
10	1	0.67291	0.01098
10	10	3.19353	0.01466
20	0	0.02968	0.00113
20	0.1	0.05557	0.00335
20	1	0.83735	0.01867
20	10	3.77534	0.02419

The RMS radius and the cell average in Fig. 5.3 compare the same curves after integration or averaging.

C.2 Spatial resolution

Bisection is used to find μ_0 , and the density is integrated with the trapezoidal rule. The iteration stops when the relative particle number error is below 10^{-10} and the density at the grid edge is below $10^{-8}/a_{\text{ho}}$. The calculation is then repeated on finer grids to measure the remaining error.

Across the eight systems, refining the LDA grid eightfold changes the energy by at most $8.9 \times 10^{-3} \hbar \omega$ and the radius by at most $2.6 \times 10^{-3} a_{\text{ho}}$. Between the fourfold and eightfold

Table C.2: Forward times used in the averages. The bound b_R measures the variation of the RMS radius over the selected values and is separate from the MCSE.

N	a/a_{ho}	density $\ell\omega$	$R^2 \ell\omega$	b_R/a_{ho}
10	0	2, 4, 7	20, 30, 40, 50	0.0002248
10	0.1	2, 4, 7	20, 30, 40, 50	0.0002235
10	1	2, 4, 7	4, 6, 8, 10	0.0003162
10	10	2, 4, 7	5, 10, 20	0.0008447
20	0	2, 4, 7	20, 30, 40, 50	0.0002435
20	0.1	2, 4, 7	20, 30, 40, 50	0.0003825
20	1	2, 4, 7	20, 30, 40, 50	0.0006983
20	10	2, 4, 7	5, 10, 15	0.0003335

grids, both changes fall to 1.5×10^{-4} in their own units. Particle number errors remain below 2×10^{-9} .

The $a/a_{\text{ho}} = 10$ histograms were also rebinned. Joining bins in pairs changes the radius by at most $8.5 \times 10^{-5} a_{\text{ho}}$; groups of four change it by $4.2 \times 10^{-4} a_{\text{ho}}$. All N peaks remain.

C.3 Selected forward times

Every density average uses $\ell\omega = 2, 4, 7$. Each time is converted to a whole number of DMC steps.

The particle coordinates follow the parent maps until the chosen time ends. Only then are they placed in the density histogram. The radius needs one value per walker and can be followed for longer. Table C.2 lists the times retained for each observable and the variation of the radius across the selected range.

The first $N = 20, a/a_{\text{ho}} = 10$ calculation supplies the energy and density. Too few families survive at late times for its radius. A second calculation with the same settings supplies R^2 . Its ancestor ESS lower bound is at least 65.60 over the selected times, and its relative variation is at most 5.67×10^{-6} .

For $N = 20$ at $a/a_{\text{ho}} = 1$ and 10, only one individual radius curve is clearly flat at late times. The other curves are too noisy to establish this on their own, but their mean over five runs is stable and passes the pooled ancestry checks.

Figure C.1 shows this distinction between the noisy individual runs and the five-run means. It is used to justify the finite range reported in Table C.2, not to claim an infinite-forward-time limit.

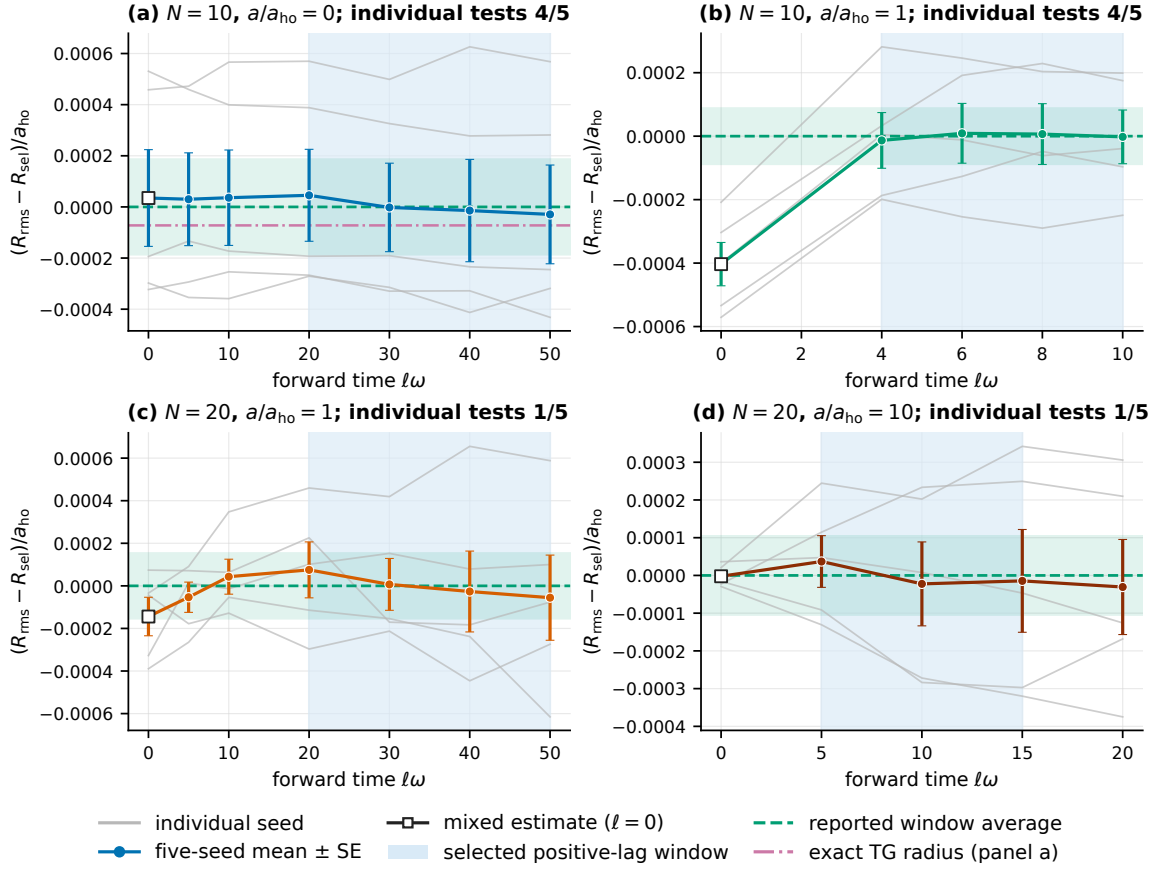


Figure C.1: RMS radius at several forward times. Gray lines show individual runs and colored points show the five-run means. Open squares are mixed estimates; blue shading marks the selected times. Panel (a) includes the exact TG radius.

The same three density times are used for all eight systems. The number of effective ancestors falls at longer times, and Fig. C.2 includes every calculated value.

The positive-time densities remain much closer to their selected mean than the mixed density, with distances well below 0.03. This supports the selected finite range but does not demonstrate convergence as $\ell \rightarrow \infty$.

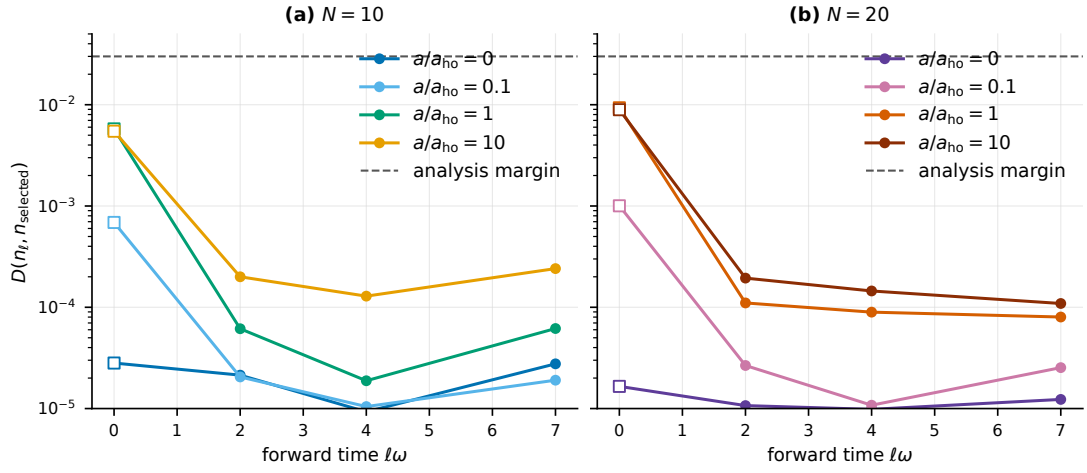


Figure C.2: Relative L^2 distance from the average over $l\omega = 2, 4, 7$. Open squares show mixed estimates and filled circles show positive forward times.

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